

11_Variance, Covariance, Correlation & Covariance Matrix

1. Variance

Definition

Variance measures **how spread out** data is from the mean.

Formula (Population)

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

Formula (Sample)

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Example (Solved)

Data:

$$X = \{2, 4, 6, 8\}$$

Step 1: Mean

$$\bar{x} = \frac{2 + 4 + 6 + 8}{4} = 5$$

Step 2: Deviations & squares

$$\sum (x_i - \bar{x})^2 = 20$$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
2	-3	9
4	-1	1
6	1	1
8	3	9

Step 3: Variance (sample)

$$s^2 = \frac{20}{4-1} = \frac{20}{3} \approx 6.67$$

2. Covariance

Definition

Covariance measures **how two variables move together**.

- Positive → move in same direction
- Negative → move in opposite directions
- Zero → no linear relationship

Formula (Sample)

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$$

Example (Solved)

Data:

$$X = \{1, 2, 3\}, Y = \{2, 4, 6\}$$

Step 1: Means

$$\bar{x} = 2, \bar{y} = 4$$

Step 2: Table

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	Product
1	2	-1	-2	2
2	4	0	0	0
3	6	1	2	2

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 4$$

Step 3: Covariance

$$\text{Cov}(X, Y) = \frac{1 \cdot 4}{3-1} = 2$$

3. Correlation

Definition

Correlation is the **standardized version of covariance**, bounded between **-1 and +1**.

Formula

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Example (Solved)

From previous example:

- $\text{Cov}(X, Y) = 2$
- $\sigma_X = 1$
- $\sigma_Y = 2$

$$r = \frac{2}{1 \times 2} = 1$$

✅ **Perfect positive correlation**

Example 01 : Find the **correlation coefficient** between:

$$X = \{1, 2, 3, 4\}$$

$$Y = \{4, 6, 8, 10\}$$

Step 1: Means

$$\bar{x} = \frac{1 + 2 + 3 + 4}{4} = 2.5$$

$$\bar{y} = \frac{4 + 6 + 8 + 10}{4} = 7$$

Step 2: Deviations and products

x	y	x - $\bar{x}$	y - $\bar{y}$	(x - $\bar{x}$)(y - $\bar{y}$)
1	4	-1.5	-3	4.5
2	6	-0.5	-1	0.5
3	8	0.5	1	0.5
4	10	1.5	3	4.5

$$\begin{aligned}\Sigma(x - \bar{x})(y - \bar{y}) &= 10 \\ \Sigma(x - \bar{x})^2 &= 5 \\ \Sigma(y - \bar{y})^2 &= 20\end{aligned}$$

Step 3: Correlation

$$r = \frac{10}{\sqrt{5 \cdot 20}} = \frac{10}{10} = 1$$

✅ **Correlation coefficient = 1** (perfect positive correlation)

Example 02 :

Compute the **correlation coefficient** for:

$$X = \{1, 2, 3, 4\}$$

$$Y = \{10, 8, 6, 4\}$$

Solution:

Step 1: Means

$$\bar{x} = 2.5, \bar{y} = 7$$

Step 2: Deviations and products

x	y	x - $\bar{x}$	y - $\bar{y}$	(x - $\bar{x}$)(y - $\bar{y}$)
1	10	-1.5	3	-4.5
2	8	-0.5	1	-0.5
3	6	0.5	-1	-0.5
4	4	1.5	-3	-4.5

$$\begin{aligned}\Sigma(x - \bar{x})(y - \bar{y}) &= -10 \\ \Sigma(x - \bar{x})^2 &= 5 \\ \Sigma(y - \bar{y})^2 &= 20\end{aligned}$$

Step 3: Correlation

$$r = \frac{-10}{\sqrt{5 \cdot 20}} = \frac{-10}{10} = -1$$

✅ **Correlation coefficient = -1** (perfect negative correlation)

4. Covariance Matrix

Definition

A covariance matrix shows covariances **between multiple variables**.

Structure (2 variables)

$$\Sigma = \begin{bmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(Y, X) & \text{Var}(Y) \end{bmatrix}$$

Example (Solved)

Data:

$$X = \{1, 2, 3\}, Y = \{2, 4, 6\}$$

We already calculated:

- $\text{Var}(X) = 1$
- $\text{Var}(Y) = 4$
- $\text{Cov}(X, Y) = 2$

Covariance Matrix

$$\Sigma = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

1. Given data

$$X = \{1, 2, 3\}, Y = \{2, 5, 8\}$$

Number of observations: $n = 3$

2. Compute the means

$$\bar{X} = \frac{1 + 2 + 3}{3} = 2$$

$$\bar{Y} = \frac{2 + 5 + 8}{3} = 5$$

3. Compute deviations from the mean

X	$X - \bar{X}$	Y	$Y - \bar{Y}$
1	-1	2	-3
2	0	5	0
3	1	8	3

4. Compute variances and covariance

(Using **sample covariance**, divide by $n - 1 = 2$)

Variance of X

$$\text{Var}(X) = \frac{(-1)^2 + 0^2 + 1^2}{2} = \frac{2}{2} = 1$$

Variance of Y

$$\text{Var}(Y) = \frac{(-3)^2 + 0^2 + 3^2}{2} = \frac{18}{2} = 9$$

Covariance of X and Y

$$\text{Cov}(X, Y) = \frac{(-1)(-3) + 0 \cdot 0 + 1 \cdot 3}{2} = \frac{6}{2} = 3$$

5. Covariance matrix

$$\Sigma = \begin{pmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(Y, X) & \text{Var}(Y) \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 3 & 9 \end{pmatrix}$$

Final Answer

$$\begin{pmatrix} 1 & 3 \\ 3 & 9 \end{pmatrix}$$

Covariance in Python

You can calculate covariance in Python using NumPy.

To use NumPy's `cov()` function, first import NumPy and define your data:

```
import numpy as np

x = np.array([2, 4, 6])
y = np.array([5, 9, 13])
cov_matrix = np.cov(x, y, ddof=1)

print(cov_matrix)
```

5. Key Differences (Quick Summary)

Concept	Measures
Variance	Spread of one variable
Covariance	Direction of relationship
Correlation	Strength + direction
Covariance Matrix	Relationships among many variables

$$\Sigma = \frac{1}{8} \begin{bmatrix} \overset{x_1}{0.71} & \overset{x_2}{-0.43} & \overset{x_3}{0.43} & \overset{x_4}{0} \\ -0.43 & 0.46 & -0.26 & 0 \\ 0.43 & -0.26 & 0.46 & 0 \\ 0 & 0 & 0 & 0.2 \end{bmatrix} \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix}$$

Set $Y = (Y_1, Y_2)'$ with $Y_1 = X_1 + X_4$ and $Y_2 = X_2 - X_4$.
Calculate the covariance matrix of Y .

Key **Formula for Variance** of a Sum/Difference of Two Random Variables:

$$\text{Var}(A \pm B) = \text{Var}(A) + \text{Var}(B) \pm 2\text{Cov}(A, B)$$

For $Y_1 = X_1 + X_4$:

$$\text{Var}(Y_1) = \text{Var}(X_1) + \text{Var}(X_4) + 2\text{Cov}(X_1, X_4)$$

For $Y_2 = X_2 - X_4$:

$$\text{Var}(Y_2) = \text{Var}(X_2) + \text{Var}(X_4) - 2\text{Cov}(X_2, X_4)$$

$$\text{Var}(Y_1) = \text{Var}(X_1) + \text{Var}(X_4) + 2\text{Cov}(X_1, X_4)$$

$$\text{Var}(Y_1) = \frac{0.71}{8} + \frac{0.2}{8} + 2(0) = \frac{0.91}{8}$$

$$\text{Var}(Y_2) = \text{Var}(X_2) + \text{Var}(X_4) - 2\text{Cov}(X_2, X_4)$$

$$\text{Var}(Y_2) = \frac{0.46}{8} + \frac{0.2}{8} - 2(0) = \frac{0.66}{8}$$

$$\text{Cov}(Y_1, Y_2) = \text{Cov}(X_1 + X_4, X_2 - X_4)$$

$$\text{Cov}(Y_1, Y_2) = \text{Cov}(X_1, X_2) + \text{Cov}(X_1, -X_4) + \text{Cov}(X_4, X_2) + \text{Cov}(X_4, -X_4)$$

$$\text{Cov}(Y_1, Y_2) = \frac{-0.43}{8} + 0 + 0 - \frac{0.2}{8} = \frac{-0.63}{8}$$

Thus, the final covariance matrix is: $\Sigma_Y = \frac{1}{8} \begin{bmatrix} 0.91 & -0.63 \\ -0.63 & 0.66 \end{bmatrix}$